

ORIGINAL ARTICLE

Characterization of a novel, biofilm dispersing, lytic bacteriophage against drug-resistant *Enterobacter cloacae*

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Abstract

Aim: To characterize a novel bacteriophage, En5822, isolated from the environment against *Enterobacter cloacae* and exploring its application as an alternate antimicrobial.

Methods and Results: Bacteriophage was isolated from sewage sample by membrane-filtration immobilization technique. It was purified and studied for its various physical properties like microscopic structure, thermal and pH stability, latent period and burst time, antimicrobial and anti-biofilm activity as well as molecular aspects by genome sequencing and analysis. En5822 is a myovirus with relative pH and thermal stability. En5822 shows a notable reduction of host bacterial biofilm as well as planktonic cultures. Whole genome sequence analysis revealed that the En5822 genome does not contain undesirable temperate lifestyle genes, antibiotic resistance genes and toxin-encoding genes.

Conclusions: En5822 displays high lytic activity, specificity and biofilm reduction capability. It has a short latent period and high burst size that aid faster activity. Its genomic and physical attributes offer possibilities for its as an alternative antimicrobial for the treatment of drug-resistant *E. cloacae* infections.

Significance and Impact of Study: The study describes a novel, naturally virulent bacteriophage from environment capable of lysing multi-drug resistant *E. cloacae* effectively. The phage could potentially serve as an alternative strategy for treating antibiotic-resistant infections.

KEYWORDS

antimicrobials, bacteriophages, biocontrol, biofilms, Enterobacteria

INTRODUCTION

Enterobacter cloacae complex (ECC), is a leading source of gram-negative infection and is placed under the WHO's list of critical MDR pathogens as an urgent threat (WHO, 2017). ECC, comprising of *Enterobacter cloacae*, *E. asburiae*, *E. hormaechei*, *E. kobei*, *E. ludwigii* and *E. nimipressuralis* is the common cause of nosocomial infections including those in neonatal intensive care units where it is responsible for high mortality rate (Annavaiah et al., 2019; Mezzatesta et al., 2012).

Most of the species in ECC show intrinsic resistance to first-generation cephalosporins, ampicillin, amoxicillin, amoxicillin-clavulanate combination and others due to constitutive expression of AmpC beta-lactamase (Annavaiah et al., 2019). The detection of chromosomally encoded carbapenemase genes in carbapenem-resistant *Enterobacter cloacae* (CREC) is alarming (Boyd et al., 2017). *E. cloacae* among other *Enterobacter* spp. is the third most commonly isolated pathogen in nosocomial settings and human clinical specimens (Potron et al., 2013; Song et al., 2010). It is known to cause bacteraemia,

endocarditis, lower respiratory tract infections, urinary tract, and abdominal infections and septic arthritis in neonatal units (Fernandez-Baca et al., 2001).

Lytic bacteriophages hold potential as alternative antibacterials given their range of mechanisms of action, virtually infinite diversity and their ability to evolve rapidly (Lin et al., 2017; Thiel, 2004). They have been employed as antibacterials directly, in synergistic combinations with antibiotics (Torres-Barceló & Hochberg, 2016), or by expressing their lytic enzymes (Fischetti, 2005). Bacteriophages offer the advantages of low toxicity, high specificity, biofilm dispersive ability and self-proliferation in phage therapy. Additionally, the rate of bacteria gaining resistance to phages is approximately 10-fold lower than gaining resistance to antibiotics (Luria & Delbrück, 1943).

Bacteriophages against different *Enterobacter* species have been isolated and characterized previously (Jamal et al., 2019; Khawaja et al., 2016; Li et al., 2016; Manohar et al., 2018, 2019; Zhao et al., 2019). Although several *E. cloacae*-specific bacteriophages have been isolated previously, a few have been assessed for their potential as phage therapy candidates. TSE phages, MJ2, ELP140, myPSH1140, PG7 are some of the *E. cloacae*-specific phages that have been studied previously for their therapeutic efficacy (Jamal et al., 2019; Khawaja et al., 2016; Manohar et al., 2018, 2019).

In this study, we describe a novel *Enterobacter* virus En5822 infecting multi-drug resistant *E. cloacae* isolated from the environment. The various characteristics of the phage, like latent period, burst size, host range, pH and thermal stability and biofilm dispersive ability was assessed. The morphological characteristics, genome and protein profiles were also studied. The genome sequence of the phage was analysed in detail for its suitability in therapy. The physiological and genomic features of bacteriophage En5822 reveal it to be a suitable candidate for phage therapy of multi-drug resistant *E. cloacae* infections.

MATERIALS AND METHODS

All the reagents and media used in the study were procured from HiMedia Laboratories Pvt. Ltd.

Bacterial host isolation and characterization

Host bacteria were isolated from 100 ml hospital wastewater collected from Nagpur city in Central India using membrane filtration method on Simmons Citrate Agar

selective media. Of the 25 isolates obtained, 20 showed growth in culture. DNA from the 20 isolates was subjected to Randomly Amplified Polymorphic DNA PCR (RAPD) using random primer 5'-ACGATGAGCC-3'. Based on colony characteristics and RAPD, eight isolates were shortlisted and confirmed by MALDI-TOF MS-Vitek MS instrument (bioMérieux) at College of Veterinary and Animal Sciences, Maharashtra Animal and Fishery Sciences University (Nagpur, Maharashtra, India).

Antibiotic susceptibility test (AST) was conducted by Kirby Bauer disk diffusion method as per Clinical Laboratory Standard Institute guidelines (Clinical Laboratory Standard Institute, 2017; Hudzicki, 2009). The antibiotics tested were as follows (potency in µg per disc): ampicillin (10), amoxycillin (20), cefuroxime (30), cefotaxime (30), piperacillin-tazobactam (100/10), ticarcillin-clavulanic acid (75/10), ceftazidime (30), cefepime (30), imipenem (10), meropenem (10), gentamicin (10), tobramycin (10), amikacin (30), ciprofloxacin (5), levofloxacin (5) and ofloxacin (5).

Bacteriophage isolation and purification

Bacteriophage En5822 was isolated from a sewage source, using membrane-filtration immobilization technique (Ghugare et al., 2017).

For plaque assay, 1 ml filtrate was incubated with 1 ml host bacteria at 37°C for 16 h. The enriched phage lysate thus obtained was centrifuged at 8801 g for 10 min and filtered through a 0.22 µm filter membrane by syringe filtration. The filtrate was serially diluted and taken for plaque assay by the double agar overlay method. About 2 mM calcium chloride (final concentration) was added to the top agar. Single plaques observed after overnight incubation at 37°C were picked from the plate individually for purification.

Plaque purification of single phage

The phage suspension described above was subjected to five rounds of purification as described previously (Ghugare et al., 2017). Briefly, a single plaque was picked using a cut tip and released into 5 ml 1× Tryptone Soya Broth (TSB) by gentle pipetting. Fifty microlitres of OD₆₀₀ 0.4–0.6 host inoculum was added to this solution and incubated overnight in a 37°C shaker incubator. Post incubation, the solution was centrifuged and the supernatant filtered through 0.22 µm filter membrane by syringe filtration. The filtrate was further subjected to five rounds of plaque purification as described and the final filtrate was used as pure single phage suspension.

Transmission electron microscopy

The sample preparation for transmission electron microscopy (TEM) was followed as described earlier (Bozzola & Russell, 1999; Spurr, 1969). Purified phage En5822 was diluted in Phage Buffer (10 mM Tris-HCl, pH 7.5; 68 mM NaCl; 10 mM MgSO₄, 1 mM CaCl₂) to a concentration of 10⁸ plaque-forming unit (PFU) ml⁻¹. Ten microlitres of phage sample was loaded onto the carbon grid of 4–5 mm diameter with 200 mesh size. Ten microlitres of staining solution (1:1 ratio) 2% uranyl acetate + 4 mM Magnesium chloride was used for staining for 1–2 min. Post staining, excess staining was washed off using sterile water. The grid was dried at room temperature for 20–30 min prior to imaging. Imaging was done at 100,000× magnification and 120 kV using TEM Model JEM 2100 by JEOL Ltd at CSIR-Centre for Cellular and Molecular Biology, Hyderabad (Telangana, India). The images were analysed using ImageJ.

One step growth curve

Fifty millilitre of host bacterial culture was allowed to grow till OD₆₀₀ of 0.6 was achieved. The culture was centrifuged at 3824 g for 10 min and the pellet was resuspended in 0.5 ml TSB. 0.5 ml phage solution (at a multiplicity of infection of 10⁻⁸) was added and the host-phage mixture was allowed to incubate for 1 min; followed by centrifugation at 3743 g for 5 min. The supernatant was collected in a separate tube for plaque assay. The pellet was resuspended in 1 ml TSB and added as an inoculum in 100 ml TSB. One millilitre of the suspension was used for plaque assay by agar overlay method in intervals of 1 min for up to 1 h. The PFU ml⁻¹ were plotted against the corresponding time points. The latent period was calculated as the initial duration during which PFU ml⁻¹ remains unchanged. The burst size for bacteriophage En5822 was calculated by dividing the average PFU ml⁻¹ in the plateau phase by a number of infective centres (Wang, 2006).

Host range determination

The host range of bacteriophage En5822 was tested against ATCC cultures of *Escherichia coli* (25922), *Alcaligenes faecalis* (8750), *Salmonella typhimurium* (13311), *Staphylococcus aureus* (25923), *Pseudomonas aeruginosa* (35554), *Klebsiella pneumoniae* (10031) and 3–4 host isolate *E. cloacae*. The cultures of these bacteria were allowed to grow to 0.6 OD₆₀₀ and then plated using 0.6% tryptone soya agar (TSA) top agar overlaid on 2%

TSA bottom agar. Five microlitres of phage solution of titre 10⁶ PFU ml⁻¹ was used for spotting post agar solidification. The plates were then incubated at 37°C overnight (Kutter, 2009).

Thermal and pH stability

For thermal stability assay of En5822, 1 ml of phage lysate of titer 10⁶ PFU ml⁻¹ was incubated at 25, 37, 50, 60, 70, 80 and 90°C for 15, 30 and 60 min in the water bath. Post-incubation, the phage suspension was serially diluted up to 10⁻³ dilution and 10 µl of the 10⁻³ dilution was plated for plaque assay by double agar overlay method and the number of plaques were counted. The experiment was performed in triplicate and statistically analysed for the significance of difference across temperatures and time-points by analysis of variance.

For pH stability studies, 1× phosphate buffered saline (PBS) was adjusted at pH 1–14 using 1 N NaOH and 1 N HCl. Phage lysate of titre 10⁶ PFU ml⁻¹ was incubated in the range of pH buffer for 60 min. Post incubation, the phage suspension was serially diluted up to 10⁻³ dilution and 10 µl of the 10⁻³ dilution was plated for plaque assay by double agar overlay method. The number of plaques was counted and plotted and the experiment was carried out in triplicate.

Phage genomic DNA isolation

Phage DNA was isolated according to Pope et al. (2011) with modifications. Ten webbed plaque assay plates were taken and 5 ml phage buffer was added per plate. Plates were stored at 4°C overnight. Five millilitre phage buffer from all the plates was pooled giving 50 ml of phage crude lysate. The lysate was centrifuged at 7735 g for 10 min at 4°C followed by filtration through 0.22 µm membrane. The filtered lysate was treated with nuclease mix (150 mM NaCl, 0.25 mg ml⁻¹ Dnase I, 0.25 mg ml⁻¹ Rnase A, 50% Glycerol) in the ratio of 4 µl nuclease mix per millilitre lysate. The mix was incubated at 37°C for 30 min followed by 1 h at room temperature. Phages were precipitated with a solution of 3.3 M NaCl and 30% (w/v) polyethylene glycol PEG-8000 and incubated on ice for 1 h followed by centrifugation at 6512 g for 40 min. Phage pellet obtained post precipitation was resuspended in 500 µl Tris-EDTA buffer. DNA extraction using phenol:chloroform extraction method. Briefly, 250 µl each of Phenol and Chloroform was added to the buffer and centrifuged post vigorous shaking. Following centrifugation, the aqueous phase was transferred to a fresh tube. DNA was recovered by ethanol precipitation

and the pellet was resuspended in 50 μ l TE buffer. DNA was stored at -20°C until use.

Genome sequencing and analysis

DNA was sent for sequencing at a concentration of ~ 4 $\mu\text{g}/100$ μl . Quality of phage DNA and its concentration was assessed by Nanodrop Lite Spectrophotometer (Thermo Fisher Scientific) and Qubit fluorometer (Thermo Fisher Scientific) using Qubit BR assay kit (Thermo Fisher Scientific; Q32853). DNA library was prepared using 1D Native barcoding genomic DNA (EXP-NBD103) and Ligation Sequencing Kit 1D (SQK-LSK108) protocol as per the manufacturer's instructions. Sequencing was carried out using long read, MinION platform (Oxford Nanopore Technologies) at CSIR-National Chemical Laboratory, Pune. The fastq reads obtained from sequencing were trimmed using Nanopack suite of tools (De Coster et al., 2018), and all reads with a quality score of 13 and above were used for further analysis. Non-viral reads were removed, and the remaining reads were used for genome assembly using the long-read aligner, Canu. The assembled contig was annotated using PATRIC (Davis et al., 2020). Detailed method of bioinformatic analysis is given in Data S1.

The assembled genome was subjected to further bioinformatic analyses for checking the phage's suitability in therapy (Philipson et al., 2018). PHASTER tool was applied on the assembled genome for baseline gene calling and functional annotation (Arndt et al., 2016). Output from the tool was then screened for the presence of the integrase gene. For prediction of phage lifestyle, the proteome data of En5822 was subjected to PHACTS tool to determine if it resembled that of temperate phages (McNair et al., 2012).

To identify the presence of antimicrobial resistance genes in the phage genome, the assembly was subjected to tools ARG-ANNOT and ResFinder that search for putative antimicrobial resistance genes in a given sequence (Gupta et al., 2014; Kleinheinz et al., 2014).

Restriction fragment analysis of genomic DNA

Two micrograms of bacteriophage DNA was treated with 20 restriction endonucleases namely *HindIII*, *ScaI*, *SphI*, *PvuII*, *HincII*, *BamHI*, *AluI*, *HinfI*, *EcoRI*, *EcoRV*, *MluI*, *PstI*, *BglII*, *HaeIII*, *KpnI*, *XbaI*, *NotI*, *NcoI*, *SmaI* and *TaqI* following manufacturer specifications (HiMedia Laboratories Pvt. Ltd.). The resultant digest was run on 1% Agarose gel at 50 V for 2–3 h and visualized under Gel-Doc.

Phage protein isolation and sodium dodecyl sulphate-polyacrylamide gel electrophoresis

Phage protein isolation protocol was adapted and modified from Boulanger (2009). Phage was precipitated using PEG-NaCl as described earlier in "Materials and Methods" section for phage DNA isolation. The phage pellet obtained was re-suspended in 0.5 ml phage buffer. This solution was subjected to sodium dodecyl sulphate-polyacrylamide gel electrophoresis (SDS-PAGE) using Laemmli sample loading buffer (120 mM Tris-Cl pH 6.8, 4% SDS, glycerol 20%), and Tris Glycine SDS gel running buffer (25 mM Tris pH 8.8, 192 mM glycine and 0.1% SDS). The protein bands were visualized by silver staining.

Antibacterial activity of En5822 against planktonic cells and host biofilm

For anti-planktonic assay, overnight grown primary culture of host bacteria was inoculated 1:100 dilution in 30 ml TSB and incubated in 37°C shaker incubator. At OD_{600} 0.6, 10^6 phage particles were added to the culture and bacterial counts were determined by spread plate method at 0, 2, 4, 6, 8, 12 and 24 h time points. The control tube was not treated with phages and bacterial counts were determined simultaneously.

The effect of bacteriophage En5822 on static biofilm formation was assessed in a standard microtiter attachment assay (Holguín et al., 2015). Host bacterial culture at 0.6 OD_{600} was diluted 1:100 in $1\times$ TSB (titre $\sim 10^6$ colony forming unit [CFU] ml^{-1}). About 200 μl of the diluted culture and 10^3 En5822 phage particles were added per well into a 96-well polystyrene plate in triplicates. An equivalent volume of phage buffer was added in control wells. The plate was sealed with Clingwrap to avoid evaporation. Biofilm density was analysed at timepoints 0, 4, 8, 12, 24, 48, 72 h by crystal violet staining. Briefly, biofilms were washed twice with $1\times$ PBS and dried. About 250 μl 0.1% Crystal Violet solution was used for staining the biofilm for 20 min. The wells were washed with $1\times$ PBS twice. The stain was decolorized by addition of 250 μl 95% ethanol and checked for absorbance at 560 nm using a spectrophotometer (Eppendorf Biospectrometer).

The effect of bacteriophage En5822 on already formed host biofilm was assessed as described (Alves et al., 2016). A 96-well polystyrene plate with 48 h old static host biofilm was taken. The wells were washed twice with $1\times$ PBS. 10^3 En5822 phage particles diluted in phage buffer was added per well and biofilm dispersion was assessed at 0, 4, 8, 12, 24, 48, 72 h post addition by crystal violet staining as described above. Control wells contained only phage

buffer without the phages. All experiments were carried out in triplicate.

RESULTS

Host isolation and characterization

The RAPD assay detected polyclonality in 20 isolates giving rise to six different RAPD patterns (Figure S1). The isolates were distributed into six types and one representative from each type was chosen for further characterization. Culture-based differences too were studied and irrespective of RAPD type, two more isolates were chosen for further characterization. All the eight isolates shortlisted were confirmed by VITEK MS-based MALDI-TOF method (Table S1). These were identified and named as 1-1, 2-2, 2-5, 3-2, 3-4, 4-5, 5-4, 5-5. The VITEK-MS identified each isolate to be *E. cloacae* and *E. asburiae* with confidence value of 50.0 thereby unable to distinguish between the two clearly. Within the scope of this study, the isolates are considered to be *E. cloacae*.

AST of all the isolates revealed resistance against ampicillin and amoxycillin. Isolates 2-5, 3-2, 3-4 and 5-5 showed additional resistance against certain cephalosporins. The resistance profile of all isolates is described in Table 1.

One of the isolates (3-4) was randomly chosen by random sampling or lottery method for all further experiments.

Phage isolation and morphological characterization

En5822 formed clear plaques of about 4–6 mm in diameter in the double agar overlay method indicating the lytic nature of the phage (Figure 1). The plaques displayed a clear centre with a halo or a semi-transparent zone

surrounding it. TEM image of phage En5822 indicates a Myoviridae morphotype (Figure 2). It has an icosahedral head of 90 ± 0.6 nm and a tail measuring 120 ± 1.5 nm. The total phage length is nearly 210 nm.

Analysis of phage growth parameters

Single step growth curve experiment for bacteriophage En5822 gave a triphasic curve with the latent period, rise phase and plateau phase (Figure 3). The latent period was deduced to be 13 min and burst size was calculated to be approximately 422 phage particles on an average per infected bacterial cell.

Host range of bacteriophage En5822

Host range analysis showed that only *E. cloacae* isolate was susceptible to En5822 bacteriophage as evident from the clear spots (Figure S2). En5822 bacteriophage is specific towards *E. cloacae* and infection to individual *Enterobacter* spp. in the ECC remains to be checked.

Thermal and pH stability

Bacteriophage En5822 was found to be thermally stable up to 50°C for 15 min while 30 and 60 min incubation caused a reduction in numbers by nearly 40%. A significant reduction of 90%–100% was seen in the phage numbers through 15, 30 and 60 min at 60°C (Figure 4). The phages did not survive for any of the timepoints from 70 to 90°C. The *p* value across the different temperatures and across timepoints was found to be $p < 0.005$.

The pH stability assay showed survivability of En5822 between 4 and 12 pH although they were most stable from 6 to 10 pH. The phage was inactive at pH ≤ 3 and ≥ 13 (Figure 5).

TABLE 1 Characterization of *Enterobacter* spp. isolates and antibiotic susceptibility test

S. No.	Samples	VITEK MS characterization	Resistance profile
1	1-1	<i>E. cloacae</i> / <i>E. asburiae</i>	Ampicillin, amoxycillin
2	2-2	<i>E. cloacae</i> / <i>E. asburiae</i>	Ampicillin, amoxycillin
3	2-5	<i>E. cloacae</i> / <i>E. asburiae</i>	Ampicillin, amoxycillin, cefepime
4	3-2	<i>E. cloacae</i> / <i>E. asburiae</i>	Ampicillin, amoxycillin, cefuroxime
5	3-4	<i>E. cloacae</i> / <i>E. asburiae</i>	Ampicillin, amoxycillin, cefepime
6	4-5	<i>E. cloacae</i> / <i>E. asburiae</i>	Ampicillin, amoxycillin
7	5-4	<i>E. cloacae</i> / <i>E. asburiae</i>	Ampicillin, amoxycillin
8	5-5	<i>E. cloacae</i> / <i>E. asburiae</i>	Ampicillin, amoxycillin, ciprofloxacin, ceftazidime

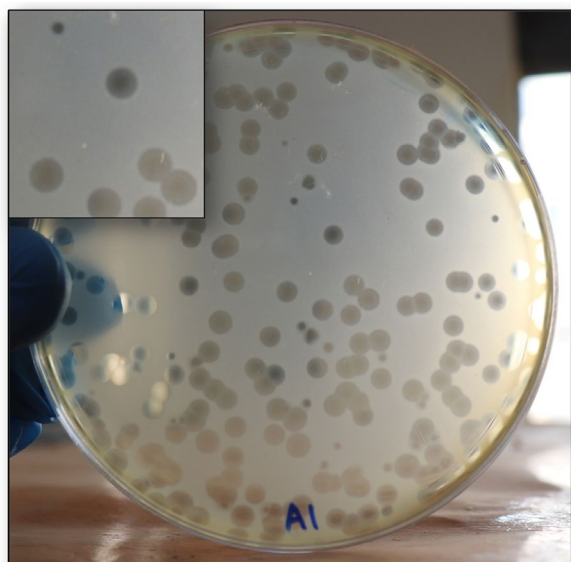


FIGURE 1 En5822 bacteriophage plaque assay plate showing distinct plaques with clear center and surrounding halo. Inset is a blown-up image of a single plaque displaying plaque characteristics

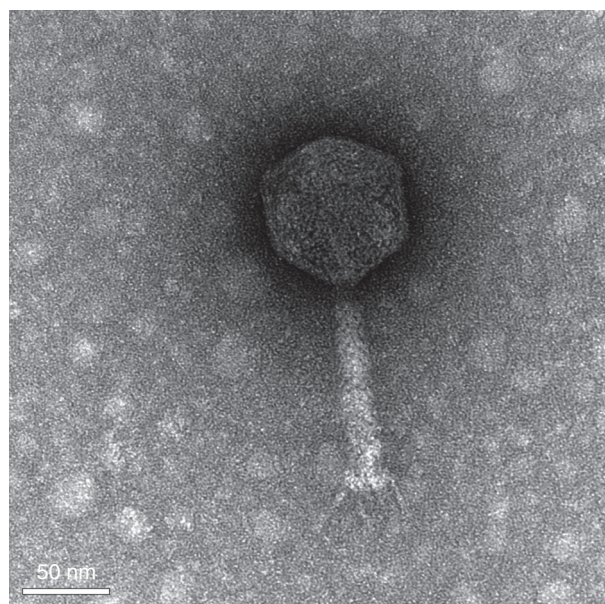


FIGURE 2 Transmission electron microscope image of En5822 bacteriophage

Genomic analysis and annotation

The genome of En5822 measures 78,128 bp with a GC content of 48.31%. Annotation identified 225 features, which included 69 phage proteins, 8 repeat regions, and 148 hypothetical proteins. Taxonomic classification of the viral reads using PATRIC indicated that 88.17% similarity to Siphoviridae bacteriophages (Figure S2).

In the taxonomic analysis using PATRIC, the reads mapped to *Salmonella* phage phSE-2 by 15%, *Escherichia* virus TLS by 4%, *Shigella* phage pSf-1 by 18.68%, *Escherichia* phage SRT8 by 14% and *Enterobacter* phage vB EcosS_NBD2 by 10% (Figure S3). A glycoside hydrolase type endolysin gene was found in duplicate at two different positions on the same contig. The PHASTER analysis did not identify integrase or repressor genes in the genome. Similarly, PHACTS Lytic Score of 0.513 was obtained which means <50% probability that phage En5822 would exhibit temperate behaviour. The results strongly suggest that the phage is likely a lytic phage.

All reads of the phage genome were profiled for antimicrobial resistance genes by ResFinder and ARG-ANNOT tool. Upon running these tools, zero hits to any antimicrobial resistance genes in the genome of En5822 were found.

Restriction fragment analysis of genomic DNA of phage En5822

The RFLP analysis of En5822 bacteriophage showed that the phage DNA remained undigested by all enzymes except *Pvu*II, *Hinc*II, *Hae*III, *Taq*I, *Alu*I and *Hin*fI. An agarose gel electrophoresis image of *Pvu*II digest of bacteriophage En5822 genome is shown in Figure 6.

Study of structural proteins of phage En5822

Protein analysis of En5822 reveals four phage protein bands of which two were major protein bands of sizes 17- and 35 kDa and two were minor bands, 25- and 40 kDa in size (Figure 7).

Antibacterial activity of En5822 against planktonic cells

In the anti-planktonic assay, viable bacterial counts start reducing 10 folds by first 2 h and then logarithmically until ~8 h (Figure 8). Beyond 8 h, the reduction in planktonic cells was comparatively lower and an overall 10^7 fold reduction in bacterial cells was seen at 24 h. Although complete elimination of host is not achieved by 24 h of the treatment with a small fraction remaining alive, more dosages before 24 h timepoint need to be added.

FIGURE 3 One step growth curve of phage En5822 showing triphasic growth pattern with latent phase, rise phase and plateau phase

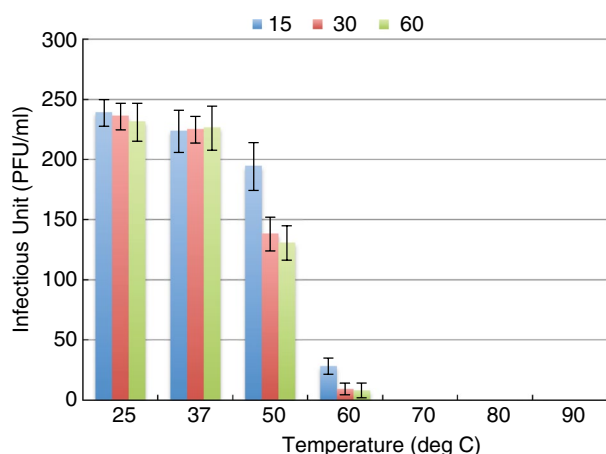
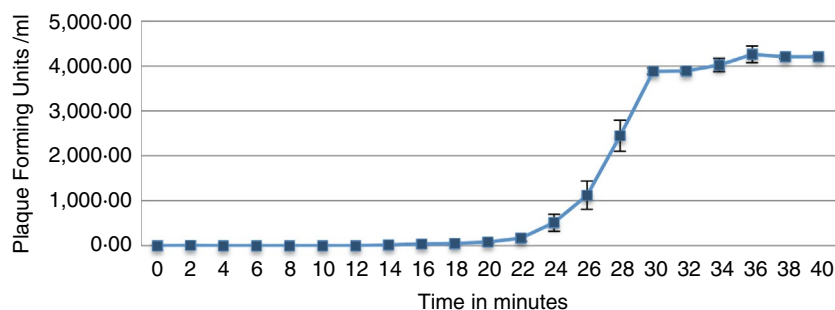


FIGURE 4 Effect of temperature on phage En5822 stability from 25 to 90°C

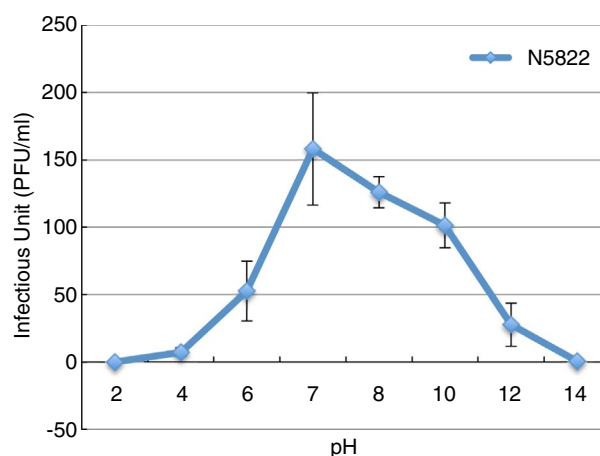


FIGURE 5 pH stability assay of En5822 showing phage stability between 7 and 10 pH

Reduction in biofilm formation by En5822

Phage En5822 reduced the formation of biofilm by ~93% in the first 4 h (Figure 9). Thereafter, by 8–12 h, the reduction ability of En5822 reduces to ~60%. Starting from 24 to 72 h again a significant reduction of ~95% is observed up till 72 h.

Dispersion of existing biofilm by En5822

The 48 h old biofilm of *E. cloacae* saw a reduction by ~72% in first 4 h of treatment with phage En5822 (Figure 10). The dispersion activity continued upto 12 h post which it reduced to ~37% and then ~55% by 24 h. The percentage reduction at 72 h is again seen to be about 76%.

DISCUSSION

Being one of the WHO Priority 1 critical pathogen having gained resistance against last-resort carbapenems among other drugs, *E. cloacae* is a serious threat that needs urgent

attention. Isolation of multi-drug resistant *Enterobacter* species from wastewater as seen in this study, especially in the vicinity of hospitals and surface water is indicative of the extent of contamination in the environment by such drug-resistant pathogens.

A large number of phages against Enterobacteriaceae group, in general, have been identified. However, specifically in the case of phages against *Enterobacter spp.*, we have come across only about 10–20 phages in EMBL-ENA database and NCBI database that have been described in detail. In the present work, we have described a lytic bacteriophage En5822, specific to *E. cloacae*.

Plaque assay studies reveal high lytic activity of En5822 bacteriophage as evident from the large clear plaques with a surrounding halo, an indication of secretion of phage-encoded lytic enzymes (Jurczak-Kurek et al., 2016). The large plaques produced by En5822 may be indicative of a larger burst size (Abedon & Culler, 2007). A short latent period and large burst size may be considered valuable to future therapeutic applications (Abedon et al., 2001; Jurczak-Kurek et al., 2016). A comparative summary of various *Enterobacter* phages with En5822 is presented in Table 2. En5822 was observed to have a higher burst size compared to most other phages.

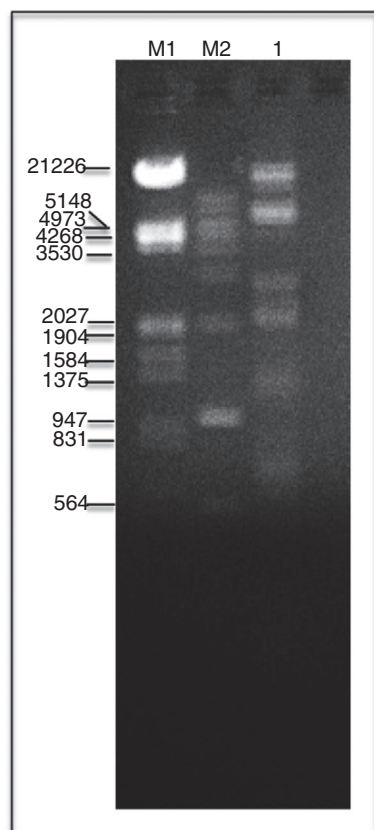


FIGURE 6 Restriction fragment analysis of En5822 genomic DNA. Lambda DNA/*EcoRI* and *HindIII* double digested molecular marker (lane 1), 1 kb molecular marker (lane 2), phage DNA digested with *PvuII* (lane 3)

The TEM results revealed that En5822 displays an icosahedral head and a contractile tail. The tail complex too is clearly visible in the TEM image revealing that En5822 is a Myovirus. Genome analysis of En5822 sequenced using Nanopore technology displays higher genomic similarities with Siphoviridae family. A comparative sequencing using long reads of Nanopore sequencing and deeper, short reads of other platforms like Illumina Seq may be helpful in understanding discrepancies, if any, in genome analysis of phage En5822.

En5822 bacteriophage host range analysis shows that it is specific to *E. cloacae* isolates. An extended host range of En5822 bacteriophage checked against individual species of ECC will be very useful. It will be interesting to understand the specificity of En5822 among various members of the complex. Since phenotypic identification of members of this group is difficult and molecular methods for distinguishing between various members too is tricky (Mezzatesta et al., 2012), phage-based species identification methods can be developed if phages like En5822 are found to exclusively infect certain member(s) of the complex.

The restriction digestion of bacteriophage En5822 genome shows relatively high resistance towards many common restriction endonucleases, indicative of a highly lytic

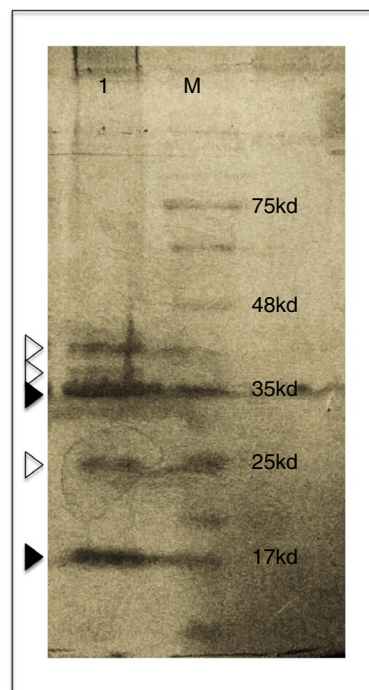


FIGURE 7 SDS-PAGE analysis of En5822 structural proteins. En5822 phage particles proteins (lane 1) and protein marker (lane 2). Major proteins indicated by solid arrowheads and minor proteins indicated by blank arrowheads. SDS-PAGE, sodium dodecyl sulphate-polyacrylamide gel electrophoresis

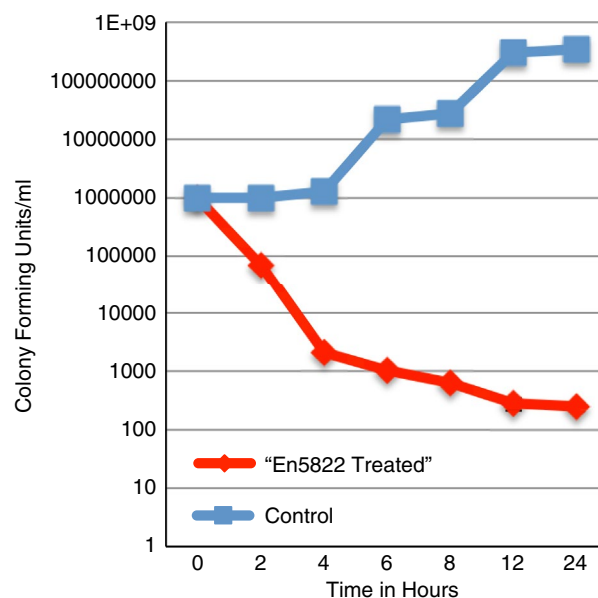


FIGURE 8 Reduction of log-phase planktonic *Enterobacter cloacae* by phage En5822 at different timepoints

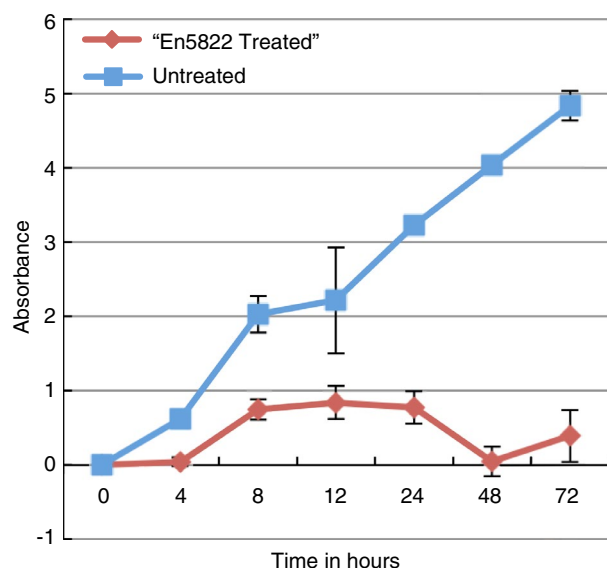


FIGURE 9 Influence of bacteriophage En5822 on formation of *Enterobacter cloacae* biofilm since 0 h timepoint. Significant reduction in biofilm formation observed 4 h onwards

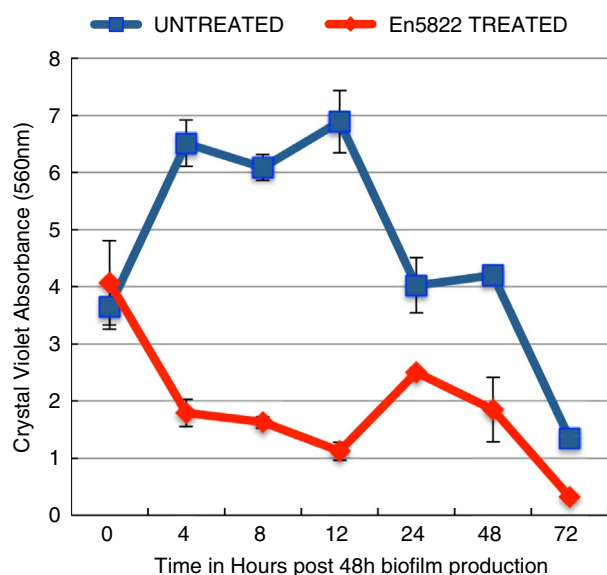


FIGURE 10 Reduction in number of 48-h-old *Enterobacter cloacae* biofilm cells after exposure to bacteriophage En5822. Biofilm reduction seen clearly within 4 h of treatment itself in static system

bacteriophage (Krüger & Bickle, 1983). It also suggests the presence of modified bases like glycosylated residues or hydroxymethyl-dUMP or genes conferring resistance to restriction as observed in the case of *P. aeruginosa* phage YuA (Ceyssens et al., 2008). Highly lytic phages are seen to acquire genomic traits that confer immunity towards various restriction enzymes for evolutionary success and fitness.

The genome of En5822 showed <20% coverage and identity with any published viral genome sequence suggesting that it is likely to be a novel virus. The glycoside hydrolase identified in the phage genome can be studied independently for its efficacy and future applications. An absence of antimicrobial resistance genes and temperate lifestyle markers indicate phage's suitability in therapeutic applications. While the PHACTS score for lytic cycle was very close to 0.5, laboratory experiments over the years did not reveal temperate behavior in bacteriophage En5822.

Coming to the actual antibacterial efficacy of phage En5822 in vitro, antiplanktonic assays show a significant reduction, although not complete killing until 24 h. It is important to note that although bacterial growth was encouraged due to the presence of growth media, the phage could still limit the bacterial growth. Optimal dosage determination of phages at time interval of 12 h may help in achieving 100% killing of the host bacteria. The efficacy of En5822 against host biofilms is promising. The biofilm formation was substantially delayed and reduced in the presence of phage even at low dosages. In the case of 48 h old biofilms, it was observed that En5822 bacteriophage showed significant dispersive action for up to 12 h, and after that, a gradual reduction in biofilm dispersive ability was noticed. Higher phage dosages for biofilm dispersion may be tried. It was noticed that the biofilm by itself reduces in both test and control 72 h timepoint which originally corresponds to 120 h (since the biofilm was already 48 h old). This effect may be due to the non-replenishment of media and sloughing off of cells. Biofilms are a menace as they provide protection to the pathogenic bacteria by forming a protective layer on any surface, including hospital instruments, catheters and most importantly, inside the human body in case of infection. They greatly increase antimicrobial resistance in the pathogen by forming a barrier to chemotherapeutics like antibiotics (Wu et al., 2015). Alternative antibacterials like En5822 bacteriophage, which effectively disperse biofilms, may prove useful in treating infections if explored further in vivo (Curtin & Donlan, 2006; Fu et al., 2010; Koo et al., 2017; Lu & Collins, 2007).

Although successful, therapeutic application of genetically engineered phages (Dedrick et al., 2019) in treating infection needs to be scrutinized carefully. A rich repertoire of bacteriophages belonging to both lysogenic and 'obligate lytic' types are present in the environment (Hobbs & Abedon, 2016). Detailed characterization of natural bacteriophages may prove extremely useful for their further development in clinical medicine. The findings of this study could expand our knowledge on applications of *Enterobacter* bacteriophages in clinical medicine.

TABLE 2 A comparison of latent periods and burst sizes of known *Enterobacter* phages

S. No.	Name of bacteriophage	Host	Latent period (min)	Burst size (PFU/cell)	Reference
1	phiEap-2	<i>Enterobacter aerogenes</i>	25	100	Li et al. (2016)
2	TSE 1	<i>E. cloacae</i>	20	360	Khawaja et al. (2016)
3	TSE 2	<i>E. cloacae</i>	20	270	Khawaja et al. (2016)
4	TSE 3	<i>E. cloacae</i>	20	311	Khawaja et al. (2016)
5	MJ2	<i>E. cloacae</i> , <i>Achromobacter xylosoxidans</i> , <i>Klebsiella pneumonia</i> , <i>Pseudomonas aeruginosa</i>	21	350	Jamal et al. (2019)
6	myPSH1140	<i>E. cloacae</i>	11	135	Manohar et al. (2019)
7	ELP140	<i>E. cloacae</i>	11	135	Manohar et al. (2018)
8	vB_EaeM_φEap-3	<i>E. aerogenes</i>	10	109	Zhao et al. (2019)
9	En5822	<i>E. cloacae</i>	13	422	Present study

Abbreviation: PFU, plaque forming unit.

Bold font/values indicates phage characterized in present study.

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CONFLICT OF INTEREST

The authors declare no conflict of interest.

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